

Major contributors to this report

- A. Srinivasa Moorthy, director, design engineering, D4X Technologies Pvt Ltd
- L. Sampath, managing director, EMST Marketing Pvt Ltd
- Shailendra Mathur, country manager, ASYS Group
- Suresh Nair, director, Leaptech Corp.

there are many advantages of using this equipment, such as lower manufacturing costs, higher component density, as well as simpler and faster automation leading to smart factories. Here is a collation of their views.

Focus on precision. Technological advancements in manufacturing will largely focus on bringing precision into the manufacturing process to reduce defects, while optimising the use of assembly operations to improve productivity.

Demand for one-stop solutions. There is an increased demand for integrated solutions to enable fully-automated control solutions. Therefore customers are shifting from using individual SMT machines to integrated one-stop SMT assembly lines to get a competitive advantage.

Shift to 0201 geometry. One of the key trends is that, India has moved straight to 0201 geometry, leapfrogging the phase of making larger SMT products.

Affordable equipment for small products. Low-cost SMT equipment lines for small products are becoming affordable. This trend is already driving the market well.

Demand for small things. With rapid increase in the use of smaller IoT devices (known as things), high volumes needed will drive the SMT market. Moreover, almost all component vendors have migrated their processes to SMT, to increase manufacturing speed, use advanced complementary metal oxide semiconductor (CMOS) technology and due to decreased interconnect widths in ICs. This has essentially driven the components industry to adopt smaller packages.

Machines for smart factories

In the era of Industry 4.0, everything will be connected through the Internet and the cloud, enabling factories to function as a smart system rather

than individual parts. Hence, as an integral part of a smart system, manufacturing equipment also need to enable automation, data collection and connectivity with the following features:

Preventive maintenance. One of the main tenets of Industry 4.0 is data collection. Manufacturing equipment with appropriate sensors will collect different kinds of data, including data on how much the machine has operated, conditions it has operated in and condition of the components of the tool.

By collecting this data, machines will be able to estimate when a component needs replacement. For example, the system would alert operators to replace a bearing that has experienced a heavier load and more frequent use sooner than a less-used one. This predictive maintenance could keep machines running more efficiently and prevent downtime.

Improved utilisation. Manufacturing operations do not always use equipment as efficiently as they could, and idle time is a frequent occurrence. It is difficult to determine the reason for this idle time. However, data collection on things like line change, program stops and feed holds improve utilisation of manufacturing equipment.

Energy savings. Smart manufacturing equipment can collect data on energy usage patterns and help save energy costs. Smart meters and other innovations help energy to flow more efficiently. A smart factory could use data on energy demand to ramp up production accordingly. Thus, energy cost is low. Smart equipment can also shut down or reduce their power usage when not in use.

Preventing improper usage. Smart manufacturing equipment can also identify improper usage cases. For example, if a machine knows the amount of force or temperature it can

withstand as well as how the operator plans to use it, it can send out an alert if it detects conditions outside of that norm. This could help prevent misuse caused by human error or malfunction, which could avoid downtime as well as worker injury.

Enhanced quality assurance. Manufacturers spend a lot of time and resources on quality assurance, taking care not to let any faulty products slip through. Smart manufacturing equipment can collect this massive pool of data that smart machine tools would collect about their use. This could help them spot potential quality issues. If a machine does not operate in the expected way, it would show up in the collated data. This, in turn, would help with traceability in manufacturing and improve quality by reducing natural variations that come with human action.

Minimising the role of humans. In smart factory systems, a drastic change will happen with respect to the roles of human employees. There will be a shift from physical operation by humans to operating, monitoring and analysing data through a computer, and providing oversights to automated operations.

Moving forward

Worldwide, the electronics manufacturing equipment market and EMS industry grow hand in hand. However, the Indian EMS industry has not achieved enough maturity in comparison to other Asian countries. Most Tier 2 and Tier 3 EMS vendors still act as assemblers—they just do the job work of PCB assemblies—rather than making the entire unit or taking on the complete production of an electronic device.

Transformation of EMS players from assemblers to complete service providers will not only enhance the demand for electronics manufacturing equipment but also help EMS players get a competitive edge in the market.

Moreover, the electronics manufacturing equipment market in India is predominantly made of imported products. Local manufacturing of these equipment is negligible. This trend needs to be changed. **EFY**